# ANTIDERIVATIVE AND DEFINITION OF INTEGRAL

## Substitution Rule

① $\int (x^2+x-1)^4 (2x+1)\,dx$

$x^2+x-1 = u \Rightarrow (2x+1)\,dx = du$

$$\int (x^2+x-1)^4 (2x+1)\,dx = \int u^4\,du = \frac{u^5}{5}+c = \frac{(x^2+x-1)^5}{5}+c$$

② $\int \sqrt{2x+1}\,dx$

$2x+1 = u \Rightarrow 2\,dx = du \Rightarrow dx = \frac{1}{2}\,du$

$$\int \sqrt{2x+1}\,dx = \frac{1}{2}\int \sqrt{u}\,du = \frac{1}{2}\,\frac{u^{3/2}}{3/2}+c = \frac{1}{3}u^{3/2}+c = \frac{1}{3}\sqrt{(2x+1)^3}+c$$

③ $\int \cos(3x+1)\,dx = \frac{1}{3}\sin(3x+1)+c$

④ $\int \cos(ax+b)\,dx$

$ax+b = u \Rightarrow a\,dx = du \Rightarrow dx = \frac{1}{a}\,du$

$$\int \cos(ax+b)\,dx = \frac{1}{a}\int \cos u\,du = \frac{1}{a}\sin u + c = \frac{1}{a}\sin(ax+b)+c$$

⑤ $\int x^2 \sin(x^3)\,dx$

$x^3 = u \Rightarrow 3x^2\,dx = du \Rightarrow x^2\,dx = \frac{1}{3}\,du$

$$\int x^2 \sin(x^3)\,dx = \frac{1}{3}\int \sin u\,du = -\frac{1}{3}\cos u + c = -\frac{1}{3}\cos(x^3)+c$$

⑥ $\int x\sqrt{x^2+1}\,dx$

$x^2+1 = u \Rightarrow 2x\,dx = du \quad x\,dx = \frac{1}{2}\,du$

$$\int x\sqrt{x^2+1}\,dx = \frac{1}{2}\int \sqrt{u}\,du = \frac{1}{2}\,\frac{u^{3/2}}{3/2}+c = \frac{1}{3}u^{3/2}+c = \frac{1}{3}\sqrt{(x^2+1)^3}+c$$

⑦ $\int \frac{x\,dx}{\sqrt{x^2+1}}$

$x^2+1 = u \Rightarrow 2x\,dx = du \Rightarrow x\,dx = \frac{1}{2}\,du$

$$\int \frac{x\,dx}{\sqrt{x^2+1}} = \frac{1}{2}\int \frac{du}{\sqrt{u}} = \frac{1}{2}\int u^{-1/2}\,du = \frac{1}{\cancel{2}}\,\frac{u^{1/2}}{\cancel{1/2}} + c = \sqrt{u} + c = \sqrt{x^2+1} + c$$

⑧ $\int \frac{\ln x\,dx}{x}$

$\ln x = u \Rightarrow \frac{dx}{x} = du$

$$\int \frac{\ln x\,dx}{x} = \int u\,du = \frac{u^2}{2} + c = \frac{1}{2}(\ln x)^2 + c$$

⑨ $\int x^2 e^{x^3} dx$

$x^3 = u \Rightarrow 3x^2 dx = du \Rightarrow x^2 dx = \frac{1}{3}\,du$

$$\int x^2 e^{x^3} dx = \frac{1}{3}\int e^u\,du = \frac{1}{3}e^u + c = \frac{1}{3}e^{x^3} + c$$

⑩ $\int \tan^2 x \sec^2 x\,dx$

$\tan x = u \Rightarrow \sec^2 x\,dx = du$

$$\int \tan^2 x \sec^2 x\,dx = \int u^2\,du = \frac{u^3}{3} + c = \frac{1}{3}\tan^3 x + c$$

⑪ $\int \frac{\sec x \tan x}{\sqrt{\sec x}}\,dx$

$\sec x = u \Rightarrow \sec x \tan x\,dx = du$

$$\int \frac{\sec x \tan x}{\sqrt{\sec x}}\,dx = \int \frac{du}{\sqrt{u}} = \int u^{-1/2}\,du = \frac{u^{1/2}}{1/2} + c = 2\sqrt{\sec x} + c$$

(12) $\int \frac{1}{x^2} \cos\left(\frac{1}{x}\right) dx$

$\frac{1}{x} = u \Rightarrow \frac{-1}{x^2} dx = du \Rightarrow \frac{1}{x^2} dx = -du$

$\int \frac{1}{x^2} \cos\left(\frac{1}{x}\right) dx = -\int \cos u \, du = -\sin u + c = -\sin\left(\frac{1}{x}\right) + c$

(13) $\int x.(x-1)^9 dx$

$x-1 = u \Rightarrow x = u+1 \Rightarrow dx = du$

$\int x.(x-1)^9 dx = \int (u+1).u^9 du = \int u^{10} du + \int u^9 du = \frac{u^{11}}{11} + \frac{u^{10}}{10} + c = \frac{(x-1)^{11}}{11} + \frac{(x-1)^{10}}{10} + c$

(14) $\int e^{\cos x} \sin x \, dx$

$\cos x = u \Rightarrow -\sin x \, dx = du \Rightarrow \sin x \, dx = -du$

$\int e^{\cos x} \sin x \, dx = -\int e^u du = -e^u + c = -e^{\cos x} + c$

(15) $\int \sqrt{\sin x} \cos x \, dx$

$\sin x = u \Rightarrow \cos x \, dx = du$

$\int \sqrt{\sin x} \cos x \, dx = \int \sqrt{u} \, du = \frac{u^{3/2}}{3/2} + c = \frac{2}{3} \sqrt{\sin^3 x} + c$

(16) $\int \sqrt{1+x^2} \, x^5 \, dx$

$1+x^2 = u \Rightarrow 2x \, dx = du \qquad x \, dx = \frac{1}{2} du$

$x^2 = u-1 \Rightarrow x^4 = u^2 - 2u + 1$

$\int \sqrt{1+x^2} \, x^5 dx = \int \sqrt{1+x^2} \, x^4 \, x \, dx = \frac{1}{2} \int u^{1/2}.(u^2-2u+1) du$

$= \frac{1}{2} \int u^{5/2} du - \int u^{3/2} du + \frac{1}{2} \int u^{1/2} du$

$= \frac{1}{2} \frac{u^{7/2}}{7/2} - \frac{u^{5/2}}{5/2} + \frac{1}{2} \frac{u^{3/2}}{3/2} + c$

$= \frac{1}{7} \sqrt{(1+x^2)^7} - \frac{2}{5} \sqrt{(1+x^2)^5} + \frac{1}{3} \sqrt{(1+x^2)^3} + c$

(17) $\int \tan x \, dx$

$\cos x = u \Rightarrow -\sin x \, dx = du \Rightarrow \sin x \, dx = -du$

$\int \tan x \, dx = \int \frac{\sin x \, dx}{\cos x} = -\int \frac{du}{u} = -\ln u + c = \ln \frac{1}{u} + c = \ln(\sec x) + c$

(18) $\int \frac{x}{\sqrt{1-4x^2}} dx$

$1-4x^2 = u \Rightarrow -8x \, dx = du \Rightarrow x \, dx = -\frac{1}{8} du$

$\int \frac{x \, dx}{\sqrt{1-4x^2}} = -\frac{1}{8} \int \frac{du}{\sqrt{u}} = -\frac{1}{8} \frac{u^{1/2}}{1/2} + c = -\frac{1}{4} \sqrt{u} + c = -\frac{1}{4} \sqrt{1-4x^2} + c$

(19) $\int e^{5x} \, dx = \frac{1}{5} e^{5x} + c$

(20) $\int \frac{x}{x^2+1} dx$

$x^2+1 = u \Rightarrow 2x \, dx = du \quad x \, dx = \frac{1}{2} du$

$\int \frac{x}{x^2+1} dx = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln|u| + c = \ln \sqrt{x^2+1} + c$

(21) $\int \frac{\sin(3\ln x)}{x} dx$

$3\ln x = u \Rightarrow \frac{3dx}{x} = du \quad \frac{dx}{x} = \frac{1}{3} du$

$\int \frac{\sin(3\ln x)}{x} dx = \frac{1}{3} \int \sin u \, du = -\frac{1}{3} \cos u + c = -\frac{1}{3} \cos(3\ln x) + c$

(22) $\int e^x \sqrt{1+e^x} \, dx$

$1+e^x = u \Rightarrow e^x dx = du$

$\int e^x \sqrt{1+e^x} \, dx = \int \sqrt{u} \, du = \frac{u^{3/2}}{3/2} + c = \frac{2}{3} \sqrt{(1+e^x)^3} + c$

(23) $\int \frac{dx}{x^2+4x+5}$

$x+2 = du \Rightarrow dx = du$

$$\int \frac{dx}{x^2+4x+5} = \int \frac{dx}{(x+2)^2+1} = \int \frac{du}{1+u^2} = \arctan u + c = \arctan(x+2) + c$$

(24) $\int \frac{dx}{\sqrt{e^{2x}-1}}$

$$\int \frac{dx}{\sqrt{e^{2x}-1}} = \int \frac{dx}{e^x\sqrt{1-e^{-2x}}} = \int \frac{e^{-x}\,dx}{\sqrt{1-e^{-2x}}} \qquad e^{-x} = u \Rightarrow e^{-x}dx = -du$$

$$= \int \frac{-du}{\sqrt{1-u^2}} = -\arcsin u + c = -\arcsin(e^{-x}) + c$$

(25) $\int \frac{dx}{x(\ln x)^2}$

$\ln x = u \Rightarrow \frac{dx}{x} = du$

$$\int \frac{dx}{x(\ln x)^2} = \int \frac{du}{u^2} = -\frac{1}{u} + c = -\frac{1}{\ln x} + c$$

## Integration By Parts (L-A-P-T-U)

(27) $\int x \cdot \underbrace{\sin x\,dx}_{dv}$

$\sin x\,dx = dv \Rightarrow v = -\cos x$

$$\int x \sin x\,dx = \int x \cdot dv = x \cdot v - \int v \cdot dx = x\cdot(-\cos x) - \int(-\cos x)\,dx$$

$$= -x\cdot\cos x + \sin x + c$$

(28) $\int \ln x\,dx$

$\ln x = u \Rightarrow \frac{dx}{x} = du$

$$\int \ln x\,dx = \int u\,dx = u\cdot x - \int x\,du = \ln x \cdot x - \int x \cdot \frac{dx}{x} = x\cdot\ln x - x + c$$

(29) $\int t^2 \cdot e^t \cdot dt$

$t^2 = u \Rightarrow 2t\,dt = du$

$e^t\,dt = dv \Rightarrow e^t = v$

$\int t^2 \cdot e^t\,dt = \int u.dv = u.v - \int v.du = t^2 \cdot e^t - \int e^t \cdot 2t\,dt$ --- (1)

$\int t \cdot \underbrace{e^t\,dt}_{dv} = t.v - \int v\,dt = t.e^t - \int e^t\,dt = t.e^t - e^t$ --- (2)

(1) ve (2) den; $\int t^2 e^t dt = t^2 . e^t - 2 . \{t.e^t - e^t\} + c$

$= t^2 . e^t - 2te^t - e^t + c$ //

(30) $\int e^x \sin x\,dx$

$\sin x = u \Rightarrow \cos x\,dx = du$

$e^x dx = dv \Rightarrow e^x = v$

$\int e^x \sin x\,dx = \int u\,dv = u.v - \int v.du = e^x \sin x - \int e^x \cos x\,dx$ -- (1)

$\cos x = u \Rightarrow -\sin x\,dx = du$

$\int e^x \cos x\,dx = \int u\,dv = u.v - \int v\,du = e^x \cos x - \int -e^x \sin x\,dx$

$= e^x \cos x + \int e^x \sin x\,dx$ --- (2)

(1) ve (2) den;

$\int e^x \sin x\,dx = e^x \sin x - \{e^x \cos x + \int e^x \sin x\,dx\}$

$= e^x \sin x - e^x \cos x - \int e^x \sin x\,dx$

$2\int e^x \sin x\,dx = e^x \sin x - e^x \cos x$

$\int e^x \sin x\,dx = \frac{e^x}{2}(\sin x - \cos x) + c$ //

(31) $\int \tan^{-1}x\,dx$

$\tan^{-1}x = \arctan x = u \Rightarrow \frac{1}{1+x^2}dx = du$

20. örnek

$\int \tan^{-1}dx = \int u\,dx = u.x - \int x\,du = x.\arctan x - \int \frac{x\,dx}{1+x^2}\cdot\frac{2}{2}$

$= x.\arctan x - \frac{1}{2}\ln(1+x^2) + c$

$= x.\arctan x - \ln\sqrt{1+x^2} + c$

(32) $\int x.\tan^{-1}x\,dx$

$\tan^{-1}x = u \Rightarrow \frac{dx}{1+x^2} = du \quad , \quad x\,dx = dv \Rightarrow v = \frac{x^2}{2}$

$\int x.\tan^{-1}x\,dx = \int u\,dv = u.v - \int v.du = \frac{x^2}{2}\arctan x - \int \frac{x^2}{2}\cdot\frac{dx}{1+x^2}$ ... (1)

$\int \frac{x^2\,dx}{1+x^2} = \int\left(1 - \frac{1}{1+x^2}\right)dx = \int dx - \int\frac{dx}{1+x^2} = x - \arctan x$ ... (2)

(1) ve (2) den;

$\int x.\tan^{-1}x\,dx = \frac{1}{2}\left(x^2\arctan x - x + \arctan x\right) + c$

(33) $\int \sin^{-1}x\,dx$

$\sin^{-1}x = u \quad \frac{dx}{\sqrt{1-x^2}} = du$

$\int \sin^{-1}x\,dx = \int u\,dx = u.x - \int x\,du = x.\arcsin x - \int\frac{x\,dx}{\sqrt{1-x^2}}$ ... (1)

$1+x^2 = s \Rightarrow -2xdx = ds \Rightarrow x\,dx = -\frac{1}{2}ds$

$\int\frac{x\,dx}{\sqrt{1-x^2}} = -\frac{1}{2}\int\frac{ds}{\sqrt{s}} = -\frac{1}{\cancel{2}}\int\frac{\sqrt{s}}{1/\cancel{2}} = -\sqrt{s} = -\sqrt{1-x^2}$ ... (2)

(1) ve (2) den; $\int \sin^{-1}x\,dx = x.\arcsin x + \sqrt{1-x^2} + c$

(34) $\int \cos(\ln x)\,dx$

$\ln x = t \Rightarrow x = e^t \Rightarrow dx = e^t\,dt$

$\cos t = u \Rightarrow -\sin t\,dt = du$

$e^t\,dt = dv \Rightarrow v = e^t$

$$\int \cos(\ln x)\,dx = \int \cos t \cdot e^t\,dt = \int u \cdot dv = u \cdot v - \int v \cdot du$$

$$= \cos t \cdot e^t - \int e^t \cdot (-\sin t \cdot dt) = e^t \cos t + \int \sin t\, e^t\,dt \quad \text{--- } (1)$$

$\sin t = s \Rightarrow \cos t\,dt = ds$

$$\int \sin t\, e^t\,dt = \int s\,dv = s \cdot v - \int v\,ds = \sin t \cdot e^t - \int e^t \cos t\,dt \quad \text{--- } (2)$$

(1) ve (2) den; $\int \cos t\, e^t\,dt = e^t \cos t + \sin t\, e^t - \int e^t \cdot \cos t\,dt$

$$\int \cos t\, e^t\,dt = \frac{e^t}{2}\left(\sin t + \cos t\right) + c$$

$$\Rightarrow \int \cos(\ln x)\,dx = \int \cos t\, e^t\,dt = \frac{e^t}{2}\left(\sin t + \cos t\right) + c$$

$$= \frac{x}{2}\left(\sin(\ln x) + \cos(\ln x)\right) + c$$

**II. Yol:** $\cos(\ln x) = u \Rightarrow -\sin(\ln x)\frac{1}{x}\,dx = du$

$$\int \cos(\ln x)\,dx = \int u\,dx = u \cdot x - \int x\,du = \cos(\ln x) \cdot x - \int x \cdot \left(-\sin(\ln x)\frac{1}{x}\,dx\right)$$

$$= \cos(\ln x) \cdot x + \int \sin(\ln x)\,dx \quad \text{--- } (1)$$

$\sin(\ln x) = v \Rightarrow \cos(\ln x)\frac{1}{x}\,dx = dv$

$$\int \sin(\ln x)\,dx = \int v\,dx = v \cdot x - \int x\,dv = \sin(\ln x) \cdot x - \int x \cdot \cos(\ln x)\frac{1}{x}\,dx$$

$$= \sin(\ln x)\, x - \int \cos(\ln x)\,dx \quad \text{--- } (2)$$

(1) ve (2) den $\int \cos(\ln x)\,dx = \frac{x}{2}\left(\sin(\ln x) + \cos(\ln x)\right)$

(35) $\int a^x dx$

$a^x = u \Rightarrow a^x . \ln a \, dx = du \Rightarrow a^x dx = \frac{1}{\ln a} du$

$\int a^x dx = \frac{1}{\ln a} \int du = \frac{1}{\ln a} u + c = \frac{1}{\ln a} a^x + c$

(36) $\int e^{2x} . \sin 5x \, dx$

$e^{2x} = u \Rightarrow 2e^{2x} = du$ , $\sin 5x \, dx = dv \Rightarrow -\frac{\cos 5x}{5} = v$

$\int e^{2x} \sin 5x \, dx = \int u . dv = u.v - \int v du = -\frac{e^{2x} . \cos 5x}{5} + \frac{2}{5} \int e^{2x} \cos 5x \, dx$ — ①

$\cos 5x \, dx = ds \Rightarrow \frac{\sin 5x}{5} = s$

$\int e^{2x} \cos 5x \, dx = \int u . ds = u.s - \int s . du = \frac{e^{2x} . \sin 5x}{5} - \frac{2}{5} \int e^{2x} \sin 5x \, dx$ — ②

① ve ② den:

$\int e^{2x} \sin 5x \, dx = -\frac{e^{2x} . \cos 5x}{5} + \frac{2}{5} \left\{ \frac{e^{2x} . \sin 5x}{5} - \frac{2}{5} \int e^{2x} . \sin 5x \, dx \right\} + c$

$25 \int e^{2x} \sin 5x \, dx = -5 e^{2x} \cos 5x + 2 e^{2x} \sin 5x - 4 \int e^{2x} . \sin 5x \, dx + c$

$\int e^{2x} \sin 5x \, dx = \frac{e^{2x}}{29} . (2 . \sin 5x - 5 \cos 5x) + c$

(37) $I = 2 \int (\ln x + \ln^2 x) \, dx$

$\ln x + \ln^2 x = u \Rightarrow \frac{1}{x} (1 + 2 \ln x) dx = du$

$\int (\ln x + \ln^2 x) dx = \int u \, dx = u . x - \int x \, du = x (\ln x + \ln^2 x) - \int (1 + 2 \ln x) dx$

$= x (\ln x + \ln^2 x) - x - 2 \int \ln x \, dx$ 28. örnekten

$= x . (\ln x + \ln^2 x) - x - 2 (x . \ln x - x) + c$

$= x (\ln x + \ln^2 x - 1 - 2 \ln x + 2) + c$

$= x . (\ln^2 x - \ln x - 1) + c$

$\Rightarrow I = 2 x (\ln^2 x - \ln x - 1) + c$ //

(38) $\int \cos 3x \cdot \cos 4x\,dx$

$\cos 3x = u \Rightarrow -3\sin 3x\,dx = du, \quad \cos 4x\,dx = dv \Rightarrow v = \frac{1}{4}\sin 4x$

$$\int \cos 3x \cos 4x\,dx = \int u\,dv = u.v - \int v.\,du = \frac{1}{4}\sin 4x.\cos 3x + \frac{3}{4}\int \sin 3x.\sin 4x\,dx \quad (1)$$

$\sin 3x = t \Rightarrow 3\cos 3x\,dx = dt, \quad \sin 4x\,dx = ds \Rightarrow s = -\frac{1}{4}\cos 4x$

$$\int \sin 3x.\sin 4x\,dx = \int t.\,ds = t.s - \int s.\,dt = \frac{-1}{4}\sin 3x \cos 4x + \frac{3}{4}\int \cos 3x.\cos 4x\,dx \quad (2)$$

(1) ve (2) den;

$$\int \cos 3x.\cos 4x\,dx = \frac{1}{4}\sin 4x.\cos 3x + \frac{3}{4}\left\{-\frac{1}{4}\sin 3x.\cos 4x + \frac{3}{4}\int \cos 3x.\cos 4x\,dx\right\}$$

$$\frac{7}{16}\int \cos 3x.\cos 4x\,dx = \frac{1}{4}\sin 4x.\cos 3x - \frac{3}{16}\sin 3x.\cos 4x$$

$$\int \cos 3x.\cos 4x\,dx = \frac{4}{7}\sin 4x \cos 3x - \frac{3}{7}\sin 3x.\cos 4x + c$$

II-Yol:

$$\int \cos 3x.\cos 4x\,dx = \frac{1}{2}\int(\cos 7x + \cos x)\,dx = \frac{1}{14}\sin 7x + \frac{1}{2}\sin x + c$$

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$\sin 4x.\cos 3x = \frac{1}{2}(\sin 7x + \sin x) \qquad \sin 3x.\cos 4x = \frac{1}{2}(\sin 7x - \sin x)$

$$\Rightarrow \frac{4}{7}\sin 4x \cos 3x - \frac{3}{7}\sin 3x.\cos 4x = \frac{4}{14}(\sin 7x + \sin x) - \frac{3}{14}(\sin 7x - \sin x)$$

$$= \frac{1}{14}\sin 7x - \frac{1}{2}\sin x$$

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(39) $\int 5^{1-x}\,dx$

$5^{1-x} = u \Rightarrow 1 - x = \frac{\ln u}{\ln 5} \Rightarrow -dx = \frac{du}{u.\ln 5}$

$$\int 5^{1-x}\,dx = -\frac{1}{\ln 5}\int \cancel{u}.\frac{du}{\cancel{u}} = -\frac{u}{\ln 5} + c = -\frac{5^{1-x}}{\ln 5} + c$$